“A Lightweight AI-Integrated V2V Communication Architecture for Collision Risk Prediction

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***Abstract*—The rapid growth of intelligent transportation systems has increased the need for reliable and low-latency vehicle-to-vehicle (V2V) communication mechanisms capable of preventing road accidents caused by delayed driver response and limited situational awareness. This paper presents an AI-driven V2V communication framework designed for real-time collision prediction and early warning generation using a lightweight edge-based architecture. The proposed system employs Raspberry Pi–based vehicle nodes that exchange telemetry data such as speed, distance, and positional information through an MQTT-based communication protocol. A Time-to-Collision (TTC) model is implemented to evaluate relative vehicle motion and classify risk levels dynamically, enabling timely hazard detection. In addition, onboard AI-based pedestrian detection enhances safety by identifying vulnerable road users and issuing proactive alerts. Experimental evaluation demonstrates low communication latency and stable real-time data exchange suitable for safety-critical applications. Performance analysis shows that the proposed system effectively identifies unsafe driving scenarios and reduces collision risk through predictive alert mechanisms. The presented framework highlights the feasibility of low-cost edge-computing platforms for intelligent vehicular safety applications and provides a scalable foundation for future multi-vehicle V2V deployments.**

***Index Terms*—**Vehicle-to-Vehicle Communication (V2V), Intelligent Transportation Systems (ITS), Time-to-Collision (TTC), MQTT Protocol, Edge Computing, Collision Prediction, Artificial Intelligence, Pedestrian Detection, Raspberry Pi, Real-Time Safety Systems, Low-Latency Communication, Internet of Things (IoT).

I. INTRODUCTION

Road traffic accidents remain one of the leading causes of injury and fatalities worldwide, with a significant percentage attributed to delayed driver reaction, insufficient awareness of surrounding vehicles, and the inability to anticipate hazardous situations in real time. Conventional safety systems primarily rely on onboard sensors and driver perception, which are often limited by environmental conditions, human error, and restricted sensing range. Vehicle-to-vehicle (V2V) communication has emerged as a promising solution for improving cooperative awareness among vehicles by enabling direct exchange of safety-critical information such as speed, position, and motion characteristics.

Recent advancements in intelligent transportation systems have focused on integrating communication technologies with edge computing and artificial intelligence to enable predictive safety mechanisms. Traditional V2V implementations commonly depend on Dedicated Short-Range Communication (DSRC) or cellular-based V2X infrastructures, which may increase deployment complexity and cost, particularly in early-stage experimental environments. Alternatively, lightweight communication protocols such as Message Queuing Telemetry Transport (MQTT) offer efficient data exchange suitable for embedded and resource-constrained platforms, making them attractive for rapid prototyping and research-oriented safety systems.

This work proposes an AI-driven V2V communication system that combines MQTT-based telemetry exchange with a Time-to-Collision (TTC) prediction model to identify potentially dangerous vehicle interactions at an early stage. Each vehicle node operates as an edge-processing unit capable of computing collision risk locally while simultaneously sharing real-time information with nearby vehicles. The integration of AI-based pedestrian detection further enhances system awareness by addressing interactions between vehicles and vulnerable road users. Unlike conventional approaches that rely heavily on infrastructure support, the proposed framework emphasizes decentralized communication and low-cost implementation using embedded platforms.

The main contributions of this work are summarized as follows:

1. Development of a lightweight V2V communication architecture using MQTT for low-latency telemetry exchange.
2. Implementation of a TTC-based collision prediction mechanism for dynamic risk classification.
3. Integration of AI-based pedestrian detection for improved road safety awareness.
4. Experimental validation demonstrating reliable real-time performance and effective collision-risk identification.

II. RELATED WORKS

Vehicle-to-Vehicle (V2V) communication has emerged as a fundamental component of intelligent transportation systems aimed at improving road safety through cooperative awareness and real-time information exchange. Early research in connected vehicular systems primarily relied on Dedicated Short-Range Communication (DSRC) and IEEE 802.11p standards to enable low-latency communication between vehicles. These systems demonstrated significant improvements in collision avoidance and traffic efficiency by allowing vehicles to exchange speed, position, and motion-related information. However, infrastructure dependency, deployment cost, and hardware complexity remain major challenges for large-scale implementation, especially in experimental and low-cost research environments.

Recent studies have explored alternative communication approaches using Internet of Things (IoT) technologies to enable flexible and scalable vehicular communication frameworks. Lightweight messaging protocols such as Message Queuing Telemetry Transport (MQTT) have gained attention due to them publish–subscribe architecture, reduced bandwidth consumption, and suitability for embedded platforms. Several researchers have demonstrated that MQTT-based communication can achieve reliable telemetry exchange with low communication overhead, making it a viable solution for prototype-level V2V and Vehicle-to-Everything (V2X) applications. These approaches highlight the potential of software-defined vehicular communication systems for rapid development and testing.

Collision prediction mechanisms have also been widely investigated in the literature. Time-to-Collision (TTC) and relative velocity models are among the most commonly adopted techniques for evaluating risk levels in dynamic driving environments. Prior works have shown that TTC-based models provide effective early warning capability in rear-end collision scenarios due to their computational simplicity and real-time applicability. However, many existing implementations rely solely on centralized processing or infrastructure-assisted computation, which can introduce latency and reduce responsiveness in safety-critical situations.

With the advancement of edge computing, recent research has focused on processing safety-related data directly on onboard devices to reduce communication delays and improve system responsiveness. Edge-assisted V2V systems enable local decision-making while maintaining cooperative communication among vehicles. Additionally, the integration of artificial intelligence techniques, particularly for pedestrian and object detection, has enhanced situational awareness beyond vehicle-only interaction models. AI-enabled perception systems allow vehicles to identify vulnerable road users and generate proactive safety alerts.

Despite these advancements, several limitations remain in existing works. Many proposed systems either focus exclusively on communication performance without integrating predictive safety intelligence or rely on expensive automotive-grade hardware platforms. Furthermore, experimental studies involving low-cost embedded systems with integrated AI-based awareness and real-time collision prediction remain limited.

To address these gaps, this work proposes an AI-driven V2V communication framework that combines MQTT-based telemetry exchange, edge-based TTC computation, and onboard AI-assisted pedestrian detection within a unified architecture. The proposed system emphasizes low-cost implementation, decentralized processing, and real-time safety response, thereby providing a practical and scalable approach for intelligent vehicular safety research.

III. SYSTEM ARCHITECTURE

The proposed AI-driven Vehicle-to-Vehicle (V2V) communication system is designed as a lightweight edge-based architecture that enables real-time data exchange and collision prediction between vehicle nodes. The system consists of two primary hardware units: a Raspberry Pi, which functions as the main processing and communication controller, and a Raspberry Pi Pico, which performs data acquisition and low-level sensor interfacing. This hybrid architecture allows efficient separation of sensing, communication, and decision-making tasks while maintaining low computational overhead.

A. Hardware Architecture

The Raspberry Pi serves as the central processing unit responsible for communication management, collision prediction, and AI-based processing. It operates as an edge-computing node that receives telemetry data, performs Time-to-Collision (TTC) calculations, and generates safety alerts. The Raspberry Pi Pico acts as a microcontroller-based sensing unit that collects vehicle-related parameters such as speed and distance from sensors or simulated inputs. The acquired data is transmitted to the Raspberry Pi through serial communication, enabling real-time processing without overloading the main processing unit. This division of responsibilities improves system efficiency by allowing the Pico to handle time-sensitive data acquisition while the Raspberry Pi focuses on computation-intensive tasks such as communication handling and AI inference.

B. Communication Architecture

Communication between vehicle units is implemented using a lightweight publish–subscribe model based on the MQTT protocol. The Raspberry Pi publishes telemetry data including speed, distance, and timestamp information to a predefined communication topic. The receiving vehicle subscribes to this topic and processes incoming data to evaluate collision risk. This architecture enables continuous bidirectional communication between vehicles while maintaining low latency and reduced network overhead. The MQTT-based communication framework ensures scalable message exchange and supports real-time updates required for safety-critical applications. By using a decentralized communication approach, each vehicle independently evaluates risk conditions without relying on centralized infrastructure.

C. Processing and Decision Layer

The processing layer integrates telemetry analysis, collision prediction, and alert generation. The received data is used to compute relative speed and Time-to-Collision values, which are then compared against predefined safety thresholds. When unsafe conditions are detected, warning messages are generated and transmitted back to the communicating vehicle. Additionally, AI-based pedestrian detection running on the Raspberry Pi enhances environmental awareness by identifying potential risks involving vulnerable road users.

D. System Workflow

The operational workflow of the proposed system can be summarized as follows: The Raspberry Pi Pico acquires vehicle parameters such as speed and distance.

1. Sensor data is transmitted to the Raspberry Pi for processing.
2. The Raspberry Pi publishes telemetry data using MQTT.
3. The receiving vehicle node subscribes to the data and computes TTC.
4. Risk levels are classified, and alerts are generated when thresholds are exceeded.
5. Warning messages are transmitted back to the originating vehicle for driver notification.

The proposed architecture demonstrates how low-cost embedded platforms can be effectively combined to implement real-time V2V safety communication while maintaining computational efficiency and scalability for future multi-vehicle extensions.

IV. PROPOSED METHODOLOGY

The proposed methodology focuses on developing a real-time AI-driven Vehicle-to-Vehicle (V2V) safety system capable of predicting collision risks through continuous telemetry exchange and edge-based processing. The system integrates sensing, communication, prediction, and alert generation within a distributed architecture to ensure low-latency decision-making without reliance on centralized infrastructure. The overall workflow consists of telemetry acquisition, data communication, collision prediction, risk classification, and warning generation.

A. Telemetry Acquisition

Vehicle parameters such as speed and inter-vehicle distance are acquired through the Raspberry Pi Pico, which functions as the sensing and data acquisition module. The Pico collects sensor readings or simulated inputs at regular intervals and transmits the data to the Raspberry Pi through serial communication. This separation ensures reliable real-time data acquisition while minimizing processing overhead on the main computation unit. The acquired telemetry data is timestamped to enable accurate synchronization during communication and latency analysis.

B. Telemetry Communication Using MQTT

The Raspberry Pi operates as the communication controller and publishes telemetry data to an MQTT broker using a predefined topic structure. Each vehicle node subscribes to the relevant telemetry topics of nearby vehicles, enabling continuous bidirectional information exchange. The MQTT publish–subscribe model ensures lightweight communication, reduced bandwidth usage, and reliable message delivery suitable for embedded platforms. Periodic transmission of telemetry data allows receiving vehicles to maintain updated information about surrounding vehicle dynamics.

C. Time-to-Collision Based Prediction: Collision risk prediction is performed using a Time-to-Collision (TTC) model, which estimates the remaining time before a potential collision occurs based on relative vehicle motion. The TTC value is calculated using the ratio of inter-vehicle distance to relative closing speed. If the relative speed indicates that vehicles are moving apart, the TTC value is treated as infinite, indicating no collision risk.

The TTC values are categorized into multiple risk levels to enable proactive safety response:

- **TTC < 1 s → Critical**
- **1 s ≤ TTC < 2 s → High**
- **2 s ≤ TTC < 4 s → Medium**
- **TTC ≥ 4 s → Safe**

This classification enables early identification of unsafe driving scenarios and allows sufficient response time for corrective action.

D. AI-Based Pedestrian Detection

In addition to vehicle-to-vehicle interaction, the system incorporates AI-based pedestrian detection to enhance environmental awareness. The Raspberry Pi executes a lightweight computer vision model capable of identifying pedestrians in the vehicle's vicinity. When a pedestrian is detected within a predefined safety range, the system increases alert priority and generates additional warning signals. This integration extends the safety mechanism beyond vehicle-only scenarios and addresses risks involving vulnerable road users.

E. Risk Evaluation and Alert Generation

The computed TTC value and AI detection results are used to evaluate overall risk conditions. When risk thresholds are exceeded, warning messages are generated and transmitted through a dedicated MQTT alert topic. Alerts are displayed as visual warnings, enabling timely driver awareness. Bidirectional alert exchange ensures that both communicating vehicles remain informed about hazardous conditions.

F. Performance Monitoring

To evaluate system performance, parameters such as communication latency, TTC values, vehicle speed, and distance measurements are continuously logged. These data points are later used for graphical analysis and validation of system responsiveness under varying driving conditions. Low latency and consistent TTC estimation indicate the suitability of the proposed methodology for real-time vehicular safety applications.

V. MATHEMATICAL MODELING

The mathematical modeling of the proposed system focuses on quantifying collision risk using vehicle motion parameters and evaluating communication performance for real-time safety applications. The model integrates relative motion analysis, Time-to-Collision (TTC) estimation, and communication latency evaluation to support reliable risk prediction and timely warning generation.

A. Relative Motion Model

Consider two vehicles moving in the same direction, where Vehicle A represents the leading vehicle and Vehicle B represents the trailing vehicle. Let the velocities of Vehicle A and Vehicle B be denoted as VA and VB, respectively, and the inter-vehicle distance be represented by d.

The relative speed between the vehicles is defined as:

$$V_{rel} = V_b - V_A$$

where:

- Vrel= relative closing speed between vehicles,
- VB = speed of trailing vehicle,
- VA = speed of leading vehicle.

If Vrel ≤ 0, the trailing vehicle is not approaching the leading vehicle, and therefore no collision risk exists.

B. Time-to-Collision (TTC) Model

The Time-to-Collision represents the estimated time remaining before two vehicles collide if their current velocities remain constant. TTC is computed as:

$$TTC = \frac{d}{Vrel}$$

where:

- d = inter-vehicle distance,
- Vrel = relative closing speed.

If the relative speed approaches zero or becomes negative, TTC is assumed to be infinite, indicating a safe condition. TTC provides a computationally efficient metric suitable for real-time embedded implementations.

The collision risk is categorized based on TTC thresholds as follows:

- TTC<1 → Critical Risk
- 1 s ≤ TTC<2 → High Risk
- 2 s ≤ TTC<4 → Medium Risk
- TTC ≥4s → Safe Condition

This classification enables early detection of unsafe driving conditions and supports timely warning generation.

C. Safe Distance Estimation

To enhance risk interpretation, the minimum safe following distance can be approximated using the reaction-time-based model:

$$d_{safe} = V_b t_r + \frac{V^2 B}{2a}$$

where:

- tr = driver reaction time,
- a = deceleration capability of the vehicle.

If the measured distance becomes less than dsafed_{safe}dsafe , the system increases alert priority, indicating a potential collision scenario.

D. Communication Latency Model

Since real-time safety applications depend on timely information exchange, communication latency is evaluated using timestamp-based measurements. The round-trip time (RTT) is calculated as:

$$R_{TT} = t_{\mathrm{recv}} - t_{sent}$$

where:

- tsent = time at which telemetry data is transmitted,
- trecvt = time at which data is received.

Lower RTT values indicate better suitability of the communication framework for real-time V2V safety messaging. The total system delay can be expressed as:

Ttotal=Tacq+Tproc+Tcomm

where:

- Tacq= data acquisition delay,
- Tproc= processing delay,
- Tcomm = communication delay.

E. Risk Evaluation Function

The overall risk level RRR is determined as a function of TTC and detected environmental conditions:

R=f (TTC, Dp)

where Dp represents pedestrian detection status. When pedestrian presence coincides with low TTC values, the system elevates the risk classification to generate immediate warnings.

This mathematical model provides a quantitative basis for real-time collision prediction and validates the effectiveness of integrating motion analysis with low-latency communication in the proposed V2V safety framework.

VI. EXPERIMENTAL SETUP

The experimental setup was designed to evaluate the performance of the proposed AI-driven Vehicle-to-Vehicle (V2V) communication system in terms of communication latency, collision prediction capability, and real-time responsiveness. The implementation focuses on validating the feasibility of a low-cost edge-based architecture for safety-oriented vehicular communication.

A. Hardware Configuration

The experimental platform consists of two vehicle nodes representing a leading and a trailing vehicle. Each node is implemented using a Raspberry Pi and a Raspberry Pi Pico operating in a distributed configuration. The Raspberry Pi Pico is responsible for telemetry acquisition, including speed and distance parameters obtained through simulated inputs or sensor interfaces. The collected data is transmitted to the Raspberry Pi via serial communication for processing and communication tasks.

The Raspberry Pi functions as the primary processing unit, executing MQTT-based communication, Time-to-Collision (TTC) computation, AI-based pedestrian detection, and alert generation. The use of separate acquisition and processing units reduces computational load and ensures stable real-time operation.

B. Communication Environment

Communication between vehicle nodes is established using a

Wi-Fi network, with MQTT employed as the messaging protocol. A centralized MQTT broker manages message exchange between nodes using a publish–subscribe model. Telemetry data including vehicle speed, inter-vehicle distance, and timestamps are transmitted periodically at fixed intervals. Each vehicle node subscribes to incoming telemetry topics and performs local risk evaluation.

The communication setup is configured to emulate short-range V2V communication conditions typically encountered in urban driving scenarios. Network parameters such as message transmission interval and data packet size are selected to ensure continuous real-time updates while minimizing communication overhead.

C. Experimental Procedure

The system was tested under multiple driving scenarios by varying vehicle speeds and inter-vehicle distances to simulate different traffic conditions. The trailing vehicle speed was adjusted relative to the leading vehicle to create safe, moderate-risk, and high-risk scenarios. For each test case, telemetry data and system response parameters were recorded for analysis.

The experimental procedure follows these steps:

1. Telemetry data is generated by the Raspberry Pi Pico and transmitted to the Raspberry Pi.
2. The Raspberry Pi publishes telemetry data to the MQTT broker.
3. The receiving vehicle node subscribes to the telemetry topic and computes TTC values.
4. Risk levels are classified based on predefined thresholds.
5. Warning messages are generated when unsafe conditions are detected.
6. Communication latency and system response time are logged for performance evaluation.

D. Performance Metrics

The performance of the proposed system is evaluated using the following metrics:

- **Communication Latency:** Measured as the round-trip time between transmitted and received telemetry messages.
- **Time-to-Collision Accuracy:** Ability of the system to correctly identify unsafe closing scenarios.
- **Alert Response Time:** Time taken between risk detection and warning generation.
- **System Stability:** Consistency of communication and prediction under varying speed and distance conditions.

E. Data Logging and Analysis

All experimental parameters, including speed, distance, TTC values, latency, and risk mode transitions, are recorded during operation. The collected data is later used to generate performance graphs such as distance versus speed interaction, latency variation, and TTC behavior under different scenarios. These analyses validate the effectiveness of the proposed system in maintaining low-latency communication and reliable collision prediction.

VII. RESULTS AND DISCUSSION

The performance of the proposed AI-driven Vehicle-to-Vehicle (V2V) communication system was evaluated through multiple experimental scenarios involving variations in vehicle speed and inter-vehicle distance. The objective of the evaluation was to analyze communication latency, collision prediction accuracy, system responsiveness, and overall operational stability under dynamic driving conditions.

A. Distance and Speed Interaction Analysis

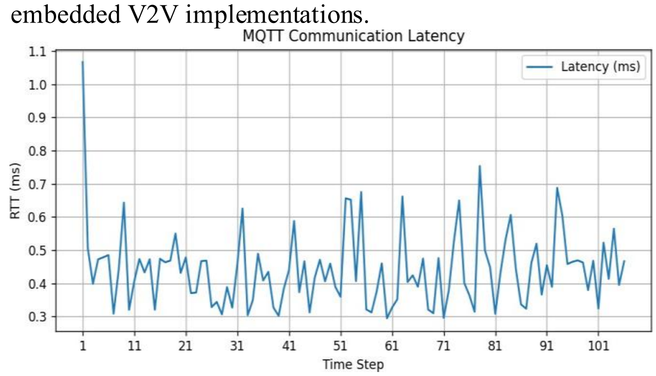
Experimental observations indicate that the relationship between vehicle speed and inter-vehicle distance plays a critical role in collision risk assessment. When the trailing vehicle operated at higher speeds relative to the leading vehicle, the Time-to-Collision (TTC) values decreased rapidly, resulting in transitions from safe mode to slow or emergency warning states. In scenarios where the inter-vehicle distance reduced below approximately 4–5 meters, the system consistently triggered high-risk or emergency alerts, demonstrating correct functioning of the TTC-based prediction mechanism.

The results confirm that the proposed algorithm responds dynamically to variations in relative velocity and maintains consistent risk classification across different operating conditions. This behavior validates the effectiveness of the TTC model in identifying unsafe following scenarios at an early stage.

B. Communication Latency Performance

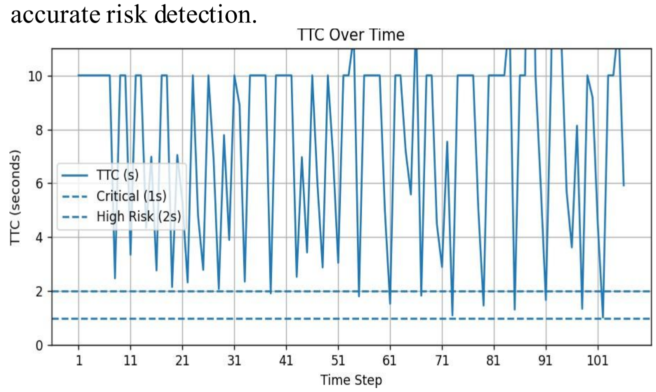
Communication latency measurements obtained during experimentation remained within a low and stable range. The observed latency values indicate that MQTT-based communication over Wi-Fi is capable of supporting real-time telemetry exchange required for safety applications. Low round-trip communication delay ensures that collision prediction calculations are performed using updated vehicle information, thereby improving system reliability.

The stability of latency values across multiple trials also demonstrates that the publish–subscribe communication architecture maintains consistent performance without significant packet delay or communication interruption. This confirms the suitability of lightweight messaging protocols for embedded V2V implementations.



C. TTC-Based Collision Prediction Performance

The TTC-based prediction model successfully identified hazardous closing conditions across different speed combinations. As the relative speed between vehicles increased, TTC values decreased proportionally, enabling earlier warning generation. Emergency alerts were generated in cases where rapid speed differences combined with short inter-vehicle distances indicated imminent collision risk.
The experimental results show that TTC provides a computationally efficient and reliable metric for real-time collision assessment, particularly in rear-end collision scenarios. The model's simplicity allows implementation on resource-constrained embedded platforms while maintaining accurate risk detection.



D. System Stability and Operational Behavior

The system demonstrated stable operation across all tested scenarios, with smooth transitions between normal, slow, fast, and emergency modes. No communication failures or inconsistent risk classifications were observed during continuous operation. The integration of Raspberry Pi and Raspberry Pi Pico enabled efficient distribution of sensing and processing tasks, reducing computational load and ensuring consistent performance.
Furthermore, the inclusion of AI-based pedestrian detection enhanced system awareness by enabling additional safety evaluation beyond vehicle-only interaction. This integration highlights the capability of combining communication-based and perception-based safety mechanisms within a single framework.

E. Discussion

The experimental findings demonstrate that low-cost embedded platforms can effectively emulate key functionalities of intelligent vehicular safety systems. The combination of MQTT-based telemetry exchange and edge-based TTC computation provides a scalable and responsive solution for real-time collision prediction. Although the current implementation focuses on a two-vehicle setup with Wi-Fi communication, the results indicate strong potential for extension toward multi-vehicle environments and advanced V2X communication technologies.
Overall, the proposed system achieves reliable collision-risk identification with low communication delay, validating the feasibility of integrating AI-assisted perception with lightweight V2V communication for intelligent transportation applications.

System Type	Communication Technology	Edge Processing	AI Integration	Latency
DSRC-based Systems [3]	IEEE 802.11p	Limited	No	Low
LTE/5G V2X Systems [12]	Cellular V2X	Partial	Limited	Moderate
Infrastructure-Assisted Models	Hybrid	Centralized	Limited	Higher
Proposed System	MQTT over Wi-Fi	Yes	Yes	Very Low

The comparison highlights that the proposed system achieves decentralized edge processing and AI integration using low-cost embedded platforms. Unlike infrastructure-dependent systems, the presented architecture provides flexible deployment with competitive latency performance, making it suitable for research and prototype-level safety applications.

VIII. CONCLUSION

This paper presented a lightweight AI-integrated Vehicle-to-Vehicle (V2V) communication architecture for real-time collision risk prediction using an edge-based processing framework. The proposed system combines MQTT-based telemetry exchange with a Time-to-Collision (TTC) prediction model and AI-based pedestrian detection to enhance road safety awareness.

Experimental evaluation demonstrated stable low-latency communication and reliable collision-risk classification under varying speed and distance conditions. The decentralized processing approach enables each vehicle node to independently evaluate risk without reliance on centralized infrastructure, improving responsiveness in safety-critical scenarios. The results confirm the feasibility of implementing intelligent vehicular safety systems using low-cost embedded platforms. Future work will focus on multi-vehicle scalability, integration with cellular V2X technologies, enhanced AI perception models, and real-world field validation under dynamic traffic environments.

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